

Unpacking Successful Literacy Interventions*

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Abstract

This paper examines experimentally the impact of a literacy intervention for primary school students in Liberia, and the main mechanisms underlying this impact. The intervention combined intensive teacher training, instructional materials, and coaching with activities aimed at increasing parental and community engagement. Using a cluster-randomized design and rich longitudinal assessment data, we find large and significant improvements in reading achievement for students in treatment schools, with gains increasing over time and reaching up to one standard deviation across key literacy domains. To examine mechanisms, we estimate a latent-factor model of skill formation that accounts for measurement error in both skills and inputs. We find that 43 percent of the total effect operates through improvements in schooling investments and teaching practices, while an additional 13 percent is accounted for by increased parental investments. We find no significant effects on numeracy skills, indicating the absence of both spillover and crowding-out effects. The results underscore the importance of school investments and the home learning environment in driving improvements in learning outcomes.

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1 Introduction

Improving learning outcomes is a central objective of education policy in low- and middle-income countries. Beyond increasing school participation, governments and development partners increasingly recognize that the accumulation of foundational skills is essential for individual welfare, labor market success, and long-run economic development. Yet, despite substantial progress in expanding access to education over the past decades, learning levels remain alarmingly low in many developing countries. Recent evidence shows that a large share of children complete primary school without acquiring basic literacy and numeracy skills, a phenomenon often referred to as the global learning crisis ([World Bank, 2018](#); [World Bank et al., 2022](#)). In response, policymakers have devoted growing attention to interventions aimed at improving the quality of instruction and strengthening the learning environment both within schools and at home.

A large literature has sought to identify policies capable of improving learning outcomes in developing countries. Randomized evaluations have documented substantial gains from remedial education programs targeted to students' learning levels ([Banerjee et al., 2007](#)), while interventions focused exclusively on expanding school inputs have often produced more modest effects ([Glewwe et al., 2009](#)). Other studies have examined the role of teacher incentives and accountability, finding that performance-based pay can improve student achievement when appropriately designed ([Muralidharan and Sundararaman, 2011](#)), whereas interventions aimed at modifying classroom practices and pedagogy have generated some of the largest and most consistent improvements in learning outcomes ([Duflo et al., 2011](#); [Piper et al., 2018](#)). More recently, reviews of this evidence have emphasized the effectiveness of interventions that directly target teaching and learning processes, particularly through structured pedagogy approaches that combine teacher training, instructional materials, and ongoing support ([Glewwe et al., 2011](#); [Evans and Popova, 2016](#)). Yet, despite the emergence of some broad patterns regarding what works to improve learning, substantial heterogeneity in program impacts remains across interventions and contexts. As a result, understanding not only whether programs work but also why they work –and through which channels they affect student achievement– is critical both for advancing our understanding of the learning process and for informing the design of education policies.

In this paper, we study the effects of the Early Grade Reading Assessment (EGRA) Plus Liberia program on children's learning outcomes and investigate the mechanisms through which these effects arise. EGRA Plus targeted grade 2 and grade 3 students and was implemented by the Liberian Ministry of Education with support from the World Bank and USAID/Liberia to improve early-grade reading skills. The intervention combined two broad components. The

first focused on improving classroom instruction through intensive teacher professional development, including regular training workshops, the provision of teaching guides and reading materials, and ongoing classroom coaching and monitoring. The second sought to strengthen the home and community learning environment by encouraging parental involvement in children's reading activities, providing parents with information on student performance, and promoting broader community engagement through reading competitions and public awareness campaigns. The program was evaluated through a cluster-randomized design that assigned schools to either a full-treatment arm, a light-treatment arm, or a control group. While schools assigned to the full-treatment arm received both components of the intervention, schools in the light-treatment arm received only the parental and community engagement activities.

Our analysis draws on data collected as part of the EGRA Plus evaluation. Participating schools were assessed at baseline, midline, and endline, with repeated cross-sections of grade 2 and grade 3 students randomly selected to participate in each assessment round. Students completed a comprehensive battery of literacy tests covering multiple dimensions of reading proficiency, such as letter recognition, phonemic awareness, reading fluency, and reading comprehension. The endline assessment also included a mathematics module designed to measure a range of foundational numeracy skills. In addition to student assessments, detailed surveys were administered to students, teachers, and school principals, collecting information on household characteristics, parental involvement in children's learning, classroom practices, teacher training and coaching, and school resources. These data provide a unique opportunity to study not only the effects of the intervention on learning outcomes but also the channels through which these effects may arise.

Previous analyses of the EGRA Plus program, conducted as part of the project's evaluation activities, documented substantial improvements in reading achievement, particularly among students exposed to the full intervention package ([Piper and Korda, 2011](#)). These analyses established the effectiveness of the intervention and informed subsequent policy discussions, but they do not investigate the mechanisms through which learning gains were generated.

We begin by revisiting the reduced-form impacts of the intervention on children's learning outcomes and potential mediating inputs. Consistent with earlier evaluation findings, we find large and statistically significant improvements in literacy achievement among students exposed to the full intervention treatment. Six months after implementation, children in full-treatment schools already exhibited higher scores across all domains of the EGRA assessment. These gains increased substantially over time. By the final assessment, students in full-treatment schools outperformed their counterparts in control schools by roughly 0.5 standard deviations in phonemic awareness and listening comprehension, 0.7 standard deviations in letter recognition, around 0.9 standard deviations in reading comprehension and familiar word fluency, and

more than one standard deviation in unfamiliar word fluency and oral reading fluency. At the same time, the intervention generated large changes in several outcomes related to parental support and classroom practices. Children in full-treatment schools were substantially more likely to report having reading books at home, being read aloud to by family members, reading aloud at home, practicing letter sounds in class, and having access to reading materials through their schools. Estimated effects on these measures range from approximately 22 to 63 percent relative to the control-group mean. In contrast, the effects of the light-treatment arm were generally smaller and largely disappeared by the end of the program.

Taken together, these findings suggest that the intervention improved literacy outcomes by changing both home and school learning environments, but they do not reveal the relative importance of these different channels. To investigate this, we develop a latent-factor model of literacy skill formation that builds on the literature on education production functions and human capital development (Todd and Wolpin, 2003; Cunha and Heckman, 2007; Cunha et al., 2010; Heckman et al., 2013; Attanasio et al., 2020). A key challenge in this context is that both learning outcomes and educational inputs are measured imperfectly through multiple survey and assessment indicators. We address this issue by jointly estimating a measurement system that combines information from a rich set of observed variables to recover latent factors capturing literacy skills, parental investments, schooling investments, household background, and household wealth. These latent factors are then embedded within a structural production-function framework that allows us to quantify both the effects of the intervention on educational inputs and the extent to which the literacy gains generated by the intervention can be attributed to improvements in different dimensions of children’s learning environments.

Our structural estimates highlight the central role of schooling investments in explaining the effectiveness of the intervention. We estimate that approximately 43 percent of the total effect of the full treatment on literacy skills operates through improvements in schooling investments, while a further 13 percent is mediated through parental investments. Together, these channels account for more than half of the overall treatment effect. By contrast, household background and wealth contribute virtually nothing to explaining the observed gains. At the same time, the direct effect of the intervention remains sizeable, accounting for roughly 44 percent of the total treatment effect. This finding suggests that important mechanisms remain outside the set of mediators captured by our model, potentially including changes in classroom management or other dimensions of the learning environment that are difficult to measure directly. Finally, applying the same framework to mathematics outcomes, which were not explicitly targeted by the intervention, we find no evidence of either positive spillovers or crowding-out effects, indicating that improvements in literacy did not come at the expense of achievement in other subjects.

The paper contributes to the literature on human capital formation and education production functions. Seminal contributions by [Todd and Wolpin \(2003\)](#) and [Cunha and Heckman \(2007\)](#) emphasize that children’s skills are produced through the interaction of multiple inputs, including parental investments, school quality, and household environments. A central challenge in empirically implementing these models is that many relevant inputs are only imperfectly observed, leading to potentially severe measurement-error problems. Recent work addresses this issue by modeling skills and investments as latent factors measured through multiple indicators and embedding these factors within structural models of skill formation ([Cunha et al., 2010](#); [Heckman et al., 2013](#); [Attanasio et al., 2020](#); [Agostinelli and Wiswall, 2025](#)). Within this literature, our paper is most closely related to [Heckman et al. \(2013\)](#) and [Attanasio et al. \(2020\)](#) in that we combine experimental variation with structural models to study the mechanisms underlying the treatment effects of educational interventions, but we differ in both setting and scope. In particular, [Heckman et al. \(2013\)](#) study the Perry Preschool Program in the US and decompose treatment effects on adult outcomes into changes in underlying personality and skill traits, highlighting how early childhood interventions operate through persistent changes in cognitive and socio-emotional skills. [Attanasio et al. \(2020\)](#) use a structural model of skill formation to analyze how an early childhood intervention in Colombia leads to improvements in children’s cognitive and socio-emotional skills by increasing parental investments. Unlike these studies, which focus on early childhood interventions and on early-life skill formation, we use a structural latent factor model of skill formation to study a randomized literacy intervention implemented during primary schooling in Liberia. This allows us to examine skill formation at school age, and to decompose learning gains into the contributions of schooling and parental investment inputs.

Finally, the paper contributes to the literature on educational interventions in developing countries. A large body of evidence has examined the effectiveness of policies aimed at improving learning outcomes, with recent reviews highlighting the success of interventions that directly target teaching and learning processes ([Evans and Popova, 2016](#); [Angrist et al., 2025](#)). The EGRA Plus program belongs to this class of interventions, combining teacher training, instructional materials, classroom support, and activities designed to foster parental engagement. Yet, while a growing consensus has emerged regarding the effectiveness of such programs, considerably less is known about the mechanisms through which they generate learning gains. By studying a large-scale literacy intervention implemented through the Liberian education system, we provide evidence on how changes in schooling practices and parental investments contribute to improvements in children’s learning outcomes.

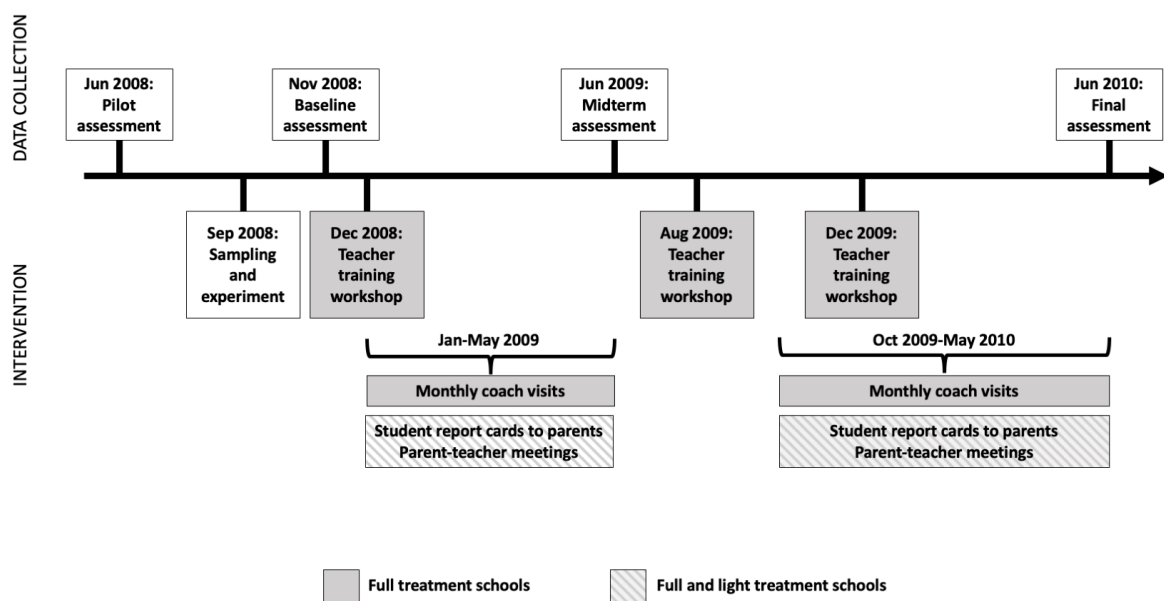
The rest of the paper is organized as follows. Section 2 provides some background on the intervention. Section 3 describes the data. Section 4 presents reduced form estimates of the effect

of the intervention on literacy skills, parental investments and schooling investments. Section 5 presents our latent factor model of skill formation and discusses the estimation strategy. Section 6 presents the estimates of the model and quantifies the importance of parental investments and schooling investments in the formation of literacy skills. Section 7 concludes.

2 Background: The EGRA Plus Liberia program

The EGRA Plus Liberia program was launched in 2008 by the Ministry of Education of Liberia, in collaboration with the World Bank and USAID/Liberia. Figure 1 provides a timeline of the key features of the intervention. Upon conducting a pilot study in June 2008, in which grade 2 and grade 3 children were examined using the EGRA assessment, reading deficiencies among Liberian children were found to be substantial. For instance, grade 2 children read an average of only 18 words per minute on the oral reading fluency test (Crouch and Poole, 2008), well below the benchmark of 70 words per minute used to identify children at risk in the same grade in the US.¹

Figure 1: Timeline of the intervention and of the data collection



Notes. The figure shows the timeline of the intervention and data collection. The intervention targeted Grade 2 and Grade 3 children from 180 public schools in Liberia, which were randomly assigned to: (i) a *full treatment* group; (ii) a *light treatment* group; (iii) a *control* group. The training workshops received by teachers in full treatment schools in December 2008, August 2009 and December 2009 lasted one week and were conducted by trained coaches. Between January and May 2009 and between October 2009 and May 2010, coaches visited full treatment schools once per month to provide feedback and support teachers with curriculum implementation or lesson planning. Student report cards were issued four times per year in both full and light treatment schools.

¹This benchmark is derived from the Dynamic Indicators of Basic Early Literacy Skills (DIBELS), an assessment format specific to the US upon which much of the EGRA is adapted. Note that the 70 words per minute benchmark reflects the standard used in 2008, when the EGRA Plus Liberia program was implemented; DIBELS benchmarks have been updated since then and are currently higher.

The poor performance of children in the pilot study prompted the launch of a two-year intervention aimed at improving the literacy skills of primary school students in second and third grade. The first component of the intervention is an intensive teacher training program. Teachers participated in workshops designed to equip them with more effective techniques for high-quality instruction in early-grade reading. Workshops lasted one week, were held once per semester and were conducted by a team of coaches. Coaches were themselves trained during week-long workshops held once per semester by an international expert in reading. In addition to the workshops, teachers were regularly visited by the coaches, typically once per month, to monitor their progress, provide feedback and support teachers with curriculum implementation or lesson planning. Finally, as part of the training intervention, teachers were given instruction manuals as well as books and other reading materials.

The second component of the intervention consists of a series of initiatives aimed at fostering parental involvement and community outreach. First, parents were encouraged to attend regular Parent-Teacher Association (PTA) meetings, during which students' performance was discussed and parents were given suggestions on how to support their children's reading activities at home. Second, teachers were asked to prepare report cards intended to inform parents about their children performance in reading.² Teachers received training on how to prepare these cards and, when necessary, were supported by coaches during their monthly school visits. Third, reading competitions among neighboring schools were organized to strengthen community outreach. Finally, four radio shows focusing on the importance of reading, current student reading levels in Liberia, and ways in which parents could help their children learn to read were broadcast.

The evaluation of the program's impact on children's literacy outcomes relies on an experiment conducted among a nationally representative sample of 180 public primary schools, selected through a two-stage procedure. In a first stage, 15 out of Liberia's 136 administrative districts were randomly selected, with probability of selection proportional to the district's population. In a second stage, three schools per district were randomly selected. Each of these schools served as the "centroid" of a corresponding cluster, which was formed by the centroid school and the three schools geographically closest to it. This resulted in a total of 45 clusters of schools. The randomization of clusters into treatment was stratified at the district level. Within each district, the three clusters were randomly assigned to one of three treatment arms: (i) a *full treatment* group (60 schools, 15 clusters) which received both the teacher training and the parental engagement components of the intervention ; (ii) a *light treatment* group (60 schools, 15 clusters) that received the parental engagement but not the teacher training activities; (iii) a

²Student report cards were issued four times per year. In addition to individual cards for each student, teachers were also asked to prepare cards showing average performance at the school level, which were discussed during PTA meetings.

control group (60 schools, 15 clusters) that received no treatment. The experimental protocol was designed at the cluster rather than at the school level for two reasons. First, for logistical and operational efficiency, the teacher training workshops were organized and delivered at the cluster level.³ Similarly, to reduce traveling times during their monthly school visits, coaches were assigned to clusters of schools. Second, and more importantly, this design ensured that all schools within the same cluster received the same treatment, thereby reducing potential spillovers across schools.

3 Data

Schools participating in the EGRA Plus intervention were assessed three times during the roll-out of the program. For each assessment, a sample of grade 2 and grade 3 students in each participating school was randomly selected to take a reading test using the EGRA Plus instrument. The sampling procedure was controlled and implemented by assessors in order to prevent teachers from selecting only the best students. On average, across schools and assessments, 18 students per school took the test. The baseline assessment was administered in November 2008, prior to the start of the intervention. The midterm assessment took place in June 2009, (approximately 6 months after the first training workshop for teachers in full-treatment schools), whereas the final assessment occurred in June 2010 (approximately 18 months after the first training workshop). An important feature of the sampling design is that it uses repeated cross-sections of grade 2 and grade 3 students rather than following the same students longitudinally.

The EGRA Plus instrument assessed students' reading skills across seven domains: letter name knowledge, phonemic awareness, familiar word fluency, unfamiliar word fluency, oral reading fluency, reading comprehension and listening comprehension. The final assessment also included a module measuring mathematics skills across six domains: number identification, quantity discrimination, missing number, addition and subtraction, multiplication and fractions. Each student completed the assessment individually with an enumerator, who recorded the student's responses and, for some domains, the time required to complete the task. The responses were subsequently converted into domain-specific test scores.⁴

Students who took the test also participated in a short survey administered by the enumerator.

³Because primary schools in Liberia are small, organizing separate workshops for each school would have required employing a large number of coaches over an extended period. Conversely, holding a single nationwide workshops for the 60 full treatment schools could have discouraged participation from teachers in more distant districts. Organizing workshops at the cluster level therefore struck a balance between achieving sufficient scale and maintaining high teacher attendance.

⁴The test scores did not affect the student's final grade at school, and enumerators ensured that children were aware of this before beginning the assessment.

The survey collected information on the student’s characteristics, family background, access to school facilities, classroom activities conducted by the teacher, and whether the child receives support at home for certain reading activities. Concurrently with the student assessment, enumerators also interviewed teachers and principals from participating schools. The teacher survey gathered information on the teachers personal and professional characteristics, the types of activities conducted in the classroom, how the curriculum was organized and lessons were planned, and the extent to which the teacher had received training or coaching visits. Data collected from principals included school resources, organization and monitoring of teachers work, and perceptions of student and parental engagement in reading during the past year.

The final sample of students includes 2,988 children from 176 schools examined in the baseline assessment; 2,882 children from 175 schools in the midterm assessment; and 2,688 children from 175 schools in the final assessment.⁵

Table 1 presents descriptive statistics from the baseline assessment, along with mean differences across treatment groups. Children in the sample are, on average, 12 years old, substantially older than the official age range for Grades 2 and 3 (approximately 7–9 years old). This pattern is consistent with broader trends in Liberia, where an estimated 75% of children are overage for their grade level ([Republic of Liberia Ministry of Education, 2016](#)). Grade repetition is also common in our sample, with 26% of children reporting having repeated at least one grade. Parental education levels are low: on average, 52% of mothers and 27% of fathers are unable to read or write. Access to reading materials is limited, with only 39% of children reporting having books at home and 21% indicating that their school provides books that can be taken home. Regarding instructional practices, nearly all students (98%) report that teachers read aloud in class; however, fewer students report exposure to more targeted literacy instruction, such as practicing letter sounds (40%) or sounding out unfamiliar words (21%).

The baseline literacy outcomes reported in panel C of Table 1 indicate that children in the sample exhibit substantial learning deficits. On average, students correctly identify 61 letters per minute and score 3.5 points on the phonemic awareness task. More strikingly, children read only about 18 words per minute in the oral reading fluency assessment and score 24% on reading comprehension. To provide context, U.S.-based DIBELS norms suggest that median oral reading fluency scores for Grade 2 and Grade 3 students range from roughly 70 to over 100 correct words per minute depending on the point in the school year. It is also worth noting that difference in literacy outcomes and in student characteristics across treatment arms are typically small and not statistically different from zero.

⁵As discussed in Section 2, the program was implemented among 180 schools. The four schools missing from the baseline assessment (three control, one full treatment) were examined in January-February 2009, after the intervention had already started, and were therefore excluded from the analysis. The additional school missing from the midterm and final assessments belonged to the control group and was excluded for similar reasons.

Table 1: Child descriptive statistics, by treatment group

	Control	Full	Light	Full– Control	Light– Control	Full– Light	Obs.
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>Panel A. Child and household sociodemographics</i>							
Male	0.54 (0.50)	0.57 (0.50)	0.54 (0.50)	0.03 (0.03)	0.01 (0.03)	0.03 (0.03)	2967
Age	12.43 (2.40)	12.64 (2.62)	12.44 (2.49)	0.21 (0.31)	0.01 (0.28)	0.19 (0.35)	2922
Grade 3	0.49 (0.50)	0.47 (0.50)	0.47 (0.50)	-0.02 (0.02)	-0.02 (0.01)	0.00 (0.02)	2963
Family speaks English at home	0.35 (0.48)	0.31 (0.46)	0.40 (0.49)	-0.03 (0.10)	0.05 (0.09)	-0.09 (0.10)	2862
Mother cannot read/write	0.48 (0.50)	0.58 (0.49)	0.51 (0.50)	0.10 (0.07)	0.03 (0.06)	0.07 (0.06)	2792
Father cannot read/write	0.27 (0.44)	0.32 (0.47)	0.23 (0.42)	0.05 (0.07)	-0.04 (0.05)	0.09 (0.06)	2771
Attended preschool	0.92 (0.27)	0.91 (0.28)	0.90 (0.30)	-0.01 (0.04)	-0.02 (0.04)	0.01 (0.05)	2934
Has ever repeated a grade	0.26 (0.44)	0.25 (0.43)	0.28 (0.45)	-0.01 (0.03)	0.02 (0.03)	-0.03 (0.04)	2939
Household asset index	0.00 (0.55)	-0.06 (0.52)	0.03 (0.54)	-0.06 (0.09)	0.03 (0.09)	-0.09 (0.09)	2945
<i>Panel B. Parental support and schooling</i>							
Has reading books at home	0.41 (0.49)	0.38 (0.49)	0.38 (0.49)	-0.03 (0.07)	-0.03 (0.05)	0.00 (0.07)	2944
Being read aloud at home	0.59 (0.49)	0.54 (0.50)	0.55 (0.50)	-0.05 (0.05)	-0.04 (0.05)	-0.00 (0.05)	2949
Reads aloud to someone at home	0.58 (0.49)	0.59 (0.49)	0.60 (0.49)	0.01 (0.05)	0.02 (0.04)	-0.01 (0.05)	2937
Has library at school	0.19 (0.39)	0.09 (0.29)	0.14 (0.35)	-0.10 (0.08)	-0.04 (0.09)	-0.05 (0.07)	2874
Has books at school to take home	0.24 (0.43)	0.16 (0.37)	0.23 (0.42)	-0.09 (0.07)	-0.02 (0.07)	-0.07 (0.06)	2929
Teacher practices letter sounds	0.41 (0.49)	0.40 (0.49)	0.39 (0.49)	-0.00 (0.09)	-0.02 (0.08)	0.01 (0.09)	2889
Reads aloud to teacher/classmates	0.83 (0.37)	0.82 (0.39)	0.84 (0.36)	-0.01 (0.04)	0.01 (0.03)	-0.02 (0.03)	2940
Teacher reads aloud	0.98 (0.15)	0.98 (0.14)	0.97 (0.16)	0.00 (0.01)	-0.00 (0.01)	0.01 (0.01)	2947
Teacher sounds unfamiliar words	0.19 (0.39)	0.19 (0.39)	0.25 (0.43)	0.00 (0.06)	0.06 (0.06)	-0.06 (0.06)	2898
<i>Panel C. Reading test scores</i>							
Correct letters per minute	60.67 (25.17)	62.35 (24.86)	60.37 (25.82)	1.68 (3.56)	-0.30 (3.24)	1.97 (3.73)	2974
Phonemic awareness	3.41 (2.32)	3.56 (2.26)	3.49 (2.30)	0.15 (0.37)	0.09 (0.30)	0.07 (0.33)	2977
Familiar word fluency	8.51 (13.54)	10.03 (14.28)	9.24 (13.86)	1.52 (1.52)	0.73 (1.72)	0.79 (1.86)	2952
Unfamiliar word fluency	1.91 (5.55)	2.51 (6.22)	2.30 (6.22)	0.60 (0.80)	0.39 (0.75)	0.21 (0.78)	2956
Oral reading fluency	18.14 (19.42)	20.83 (20.26)	19.77 (20.37)	2.70 (2.65)	1.63 (2.87)	1.06 (3.06)	2958
Reading comprehension	23.70 (23.86)	25.81 (24.37)	25.74 (24.44)	2.11 (3.55)	2.04 (3.58)	0.07 (3.72)	2958
Listening comprehension	32.64 (21.56)	33.58 (20.11)	34.51 (19.84)	0.94 (2.16)	1.87 (2.46)	-0.93 (2.23)	2988

Notes. * p<0.10, ** p<0.05, *** p<0.01. Numbers in parentheses are robust standard errors for the differences in columns (4)-(6), and standard deviations elsewhere.

4 Reduced form estimates

To estimate the causal effects of the intervention on children outcomes, we rely on the randomized assignment of schools across treatment arms and estimate reduced-form intent-to-treat specifications of the following form:

$$Y_{icd} = \alpha + \beta^F Full_c + \beta^L Light_c + X'_{icd}\gamma + \delta_d + \varepsilon_{icd} \quad (1)$$

where Y_{icd} denotes the outcome of interest for child i in schooling cluster c and district d ; $Full_c$ and $Light_c$ are indicators for assignment to the Full and Light treatment arms, respectively; X_{icd} is a vector of demographic child characteristics, including a continuous measure of age and gender and grade dummies; δ_d denotes district fixed effects, which account for the stratified randomization design at the district level; and ε_{icd} is an error term. Standard errors are clustered at the schooling cluster level, the unit of treatment assignment.

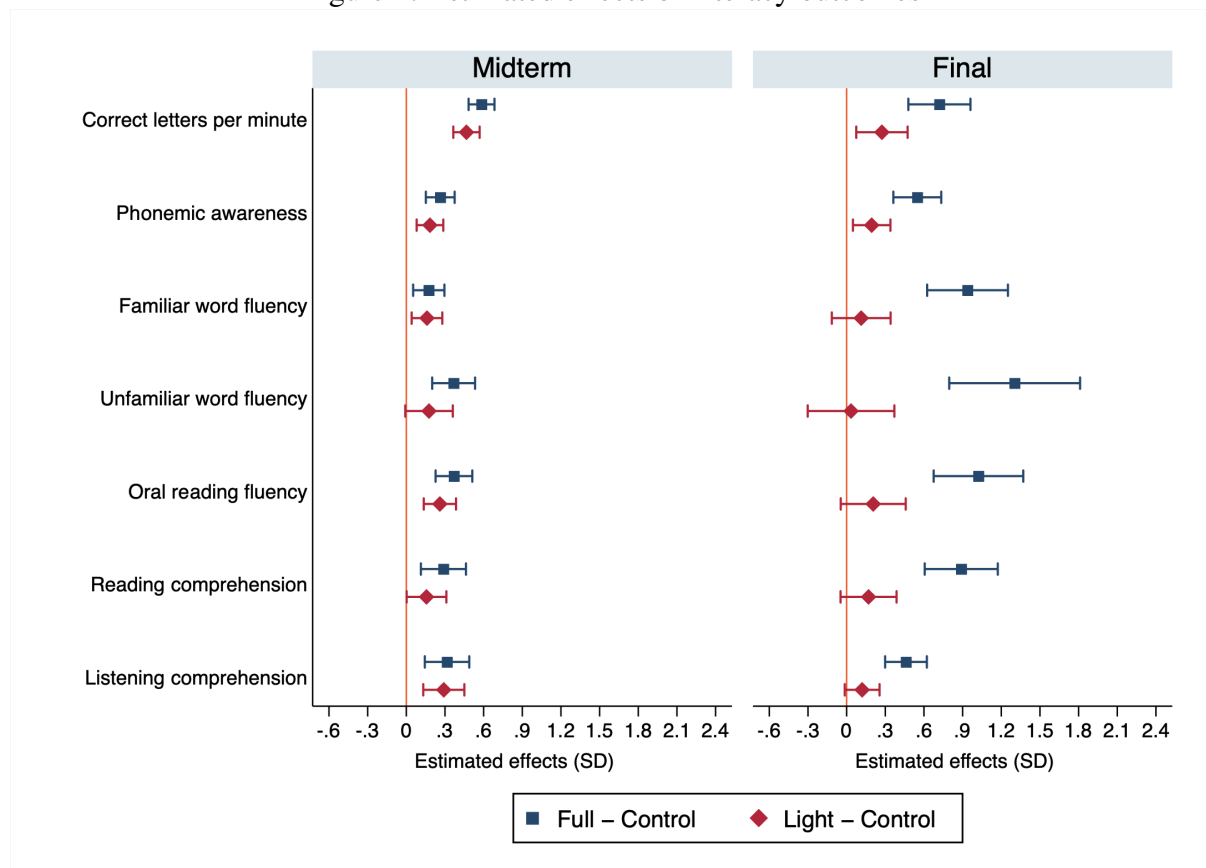
The parameters of interest are β^F and β^L , which capture, respectively, the effect of receiving the Full or Light treatment relative to the control group. Because the experimental sample consists of repeated cross-sections of students, we estimate equation (1) separately for the midterm and final assessments. To facilitate interpretation, all literacy outcomes are standardized to have mean zero and unit standard deviation in the control group within each survey round.

Figure 2 shows estimates of β^F and β^L for several literacy outcomes, corresponding to the seven domains examined in the EGRA assessment. In the midterm examination, children in full treatment schools exhibit significantly higher scores than children in control schools across all seven domains. Estimated effects are large, ranging from 0.17 standard deviations for familiar word fluency to 0.58 standard deviations for correct letters per minute. For the light treatment, estimated effects are significantly different from zero for all literacy outcomes except unfamiliar word fluency. Although the point estimates are smaller than those associated with the full treatment, the differences between the full and light treatments are statistically significant for only two outcomes (Appendix Table A1).

Program effects associated with the full treatment become even larger in the final assessment. Relative to the control group, test scores of children in full treatment schools increased by roughly 0.5 standard deviations for phonemic awareness and listening comprehension, by 0.7 standard deviations for correct letters per minute, by 0.9 standard deviations for reading comprehension and familiar word fluency, and by more than 1 standard deviation for unfamiliar word fluency and oral reading fluency. In contrast, children in light treatment schools scored significantly higher than children in control schools only in correct letters per minute and phonemic awareness, suggesting that the positive midterm effects of the light treatment in-

tervention did not persist in the final assessment. These results are robust to alternative specifications of the control variables, including estimating equation (1) without control variables and specifications that include only district fixed effects (Appendix Tables A1 and A2).

Figure 2: Estimated effects on literacy outcomes

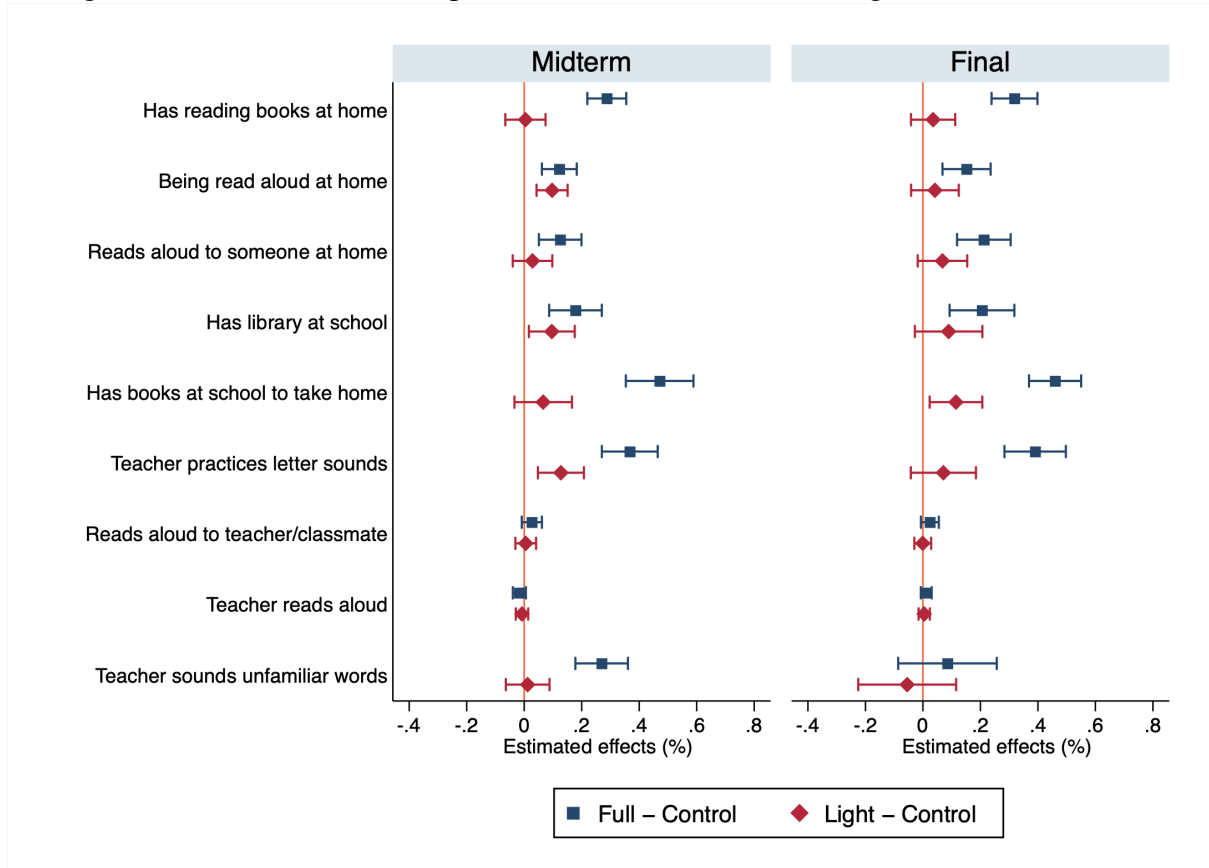


Notes. The figures show the point estimates and 90% confidence intervals of the effect of the full and light treatment on the seven reading domains examined in the EGRA assessment. Literacy outcomes are standardized to have mean zero and unit standard deviation in the control group within each survey round. Estimates are based on equation (1) and include a continuous measure of age, gender and grade dummies and district fixed effects as controls. Confidence intervals are calculated based on standard errors clustered at the schooling cluster level.

In Appendix Figure A1 and Appendix Table A3 we also look at treatment effects on the mathematics domains measured in the final assessment. While the intervention was not designed to improve mathematics achievement, children in full-treatment schools exhibit positive and statistically significant gains in a number of domains, including quantity discrimination, addition, and fractions, with estimated effects ranging from approximately 0.2 to 0.6 standard deviations. These effects are generally smaller and less pervasive than those observed for literacy outcomes, and the light treatment generates few significant improvements. Taken together, these results suggest the absence of negative spillovers from the literacy intervention to mathematics learning, but provide only limited evidence of positive spillovers across subjects.

Next, we start analyzing potential mechanisms for the estimated effects on reading test scores by looking at how the intervention affected several outcomes related to parental support, schooling, and teaching practices. Figure 3 shows that, in both the midterm and final assessments,

Figure 3: Estimated effects on parental investment and schooling investment outcomes



Notes. The figures show the point estimates and 90% confidence intervals of the effect of the full and light treatment on parental support and teaching outcomes. Estimates are based on equation (1) and include a continuous measure of age, gender and grade dummies and district fixed effects as controls. Confidence intervals are calculated based on standard errors clustered at the schooling cluster level.

children in full-treatment schools report more frequently having reading books at home, being read aloud to by someone at home, and reading aloud to someone at home. Estimated effects are large, corresponding to increases of between 22% and 63% relative to the mean in the control group. Similarly, children in full-treatment schools report having more books at school to take home and practicing certain reading activities with their teachers more frequently, such as practicing letter sounds or sounding out unfamiliar words. In contrast, estimated effects for the light treatment are generally positive but not statistically different from zero for most parental-support and teaching outcomes.

5 Modelling skill formation

To interpret the mechanisms behind the experimental results presented in the previous section, we explicitly model the relationship between literacy skills, parental investments and schooling investments within the context of a human capital production function. To fix ideas, suppose we wish to estimate a production function for child literacy skills such as

$$\theta_t = f_t(P_t, S_t, H_t, W_t, X_t, \eta_t) \quad (2)$$

where θ_t represents the child's literacy skills at time t ; P_t are parental investments; S_t are schooling investments, which include material inputs as well as teaching effort and practices; H_t is the child's household background; W_t is household wealth; X_t is a vector of child characteristics, such as age, gender and grade; and η_t is a random shock to child development. Apart from parental and schooling investments, we also include household background and wealth in equation (2) as these are typically found to be important drivers of skill formation.

An important issue in the estimation of equation (2) is the fact that both the inputs of the production function and the child's literacy skills are unobservable. In Section 4 we presented several measures that could approximate some of these factors, but these are affected by measurement error and could therefore produce biased estimates of the parameters of the production function. For this reason, we follow earlier work by [Cunha et al. \(2010\)](#), [Heckman et al. \(2013\)](#) and [Attanasio et al. \(2020\)](#) and develop a measurement system linking the observed measures to latent factors and estimate the distribution of such factors.

5.1 Factor model and measurement system

Suppose we have $\mathcal{Z}_t^\theta$ measures of child's literacy skills θ in period t . Moreover, for each input $k \in \{P, S, H, W\}$, we have $\mathcal{Z}_t^k$ measures of input k . We denote z_{jt}^θ the j -th measure of child's literacy skill at t and z_{jt}^k the j -th measure of production function input of type $k \in \{P, S, H, W\}$. We assume a dedicated measurement system, that is one in which each measure loads only on one factor ([Gorsuch, 1983, 2003](#)). Thus, the measurement system can be written as follows

$$z_{jt}^\theta = \nu_{jt}^\theta + \lambda_{jt}^\theta \theta_t + \epsilon_{jt}^\theta \quad (3)$$

$$z_{jt}^P = \nu_{jt}^P + \lambda_{jt}^P P_t + \epsilon_{jt}^P \quad (4)$$

$$z_{jt}^S = \nu_{jt}^S + \lambda_{jt}^S S_t + \epsilon_{jt}^S \quad (5)$$

$$z_{jt}^H = \nu_{jt}^H + \lambda_{jt}^H H_t + \epsilon_{jt}^H \quad (6)$$

$$z_{jt}^W = \nu_{jt}^W + \lambda_{jt}^W W_t + \epsilon_{jt}^W \quad (7)$$

where the terms $\nu_{jt}^\theta, \nu_{jt}^P, \nu_{jt}^S, \nu_{jt}^H, \nu_{jt}^W$ are intercepts, the terms $\lambda_{jt}^\theta, \lambda_{jt}^P, \lambda_{jt}^S, \lambda_{jt}^H, \lambda_{jt}^W$ are factor loadings, and the terms $\epsilon_{jt}^\theta, \epsilon_{jt}^P, \epsilon_{jt}^S, \epsilon_{jt}^H, \epsilon_{jt}^W$ are measurement error terms which are assumed independent of the latent factors and of each other. We also assume that the measurement system is invariant between control, light and full treatment, i.e. the intercepts and factor loading are the same across groups.

Table 2 shows the measures associated to each factor. Because for parental investments, schooling investments, household background and wealth the measures are discrete, we assume that each measure has a latent representation z_{jt}^{*k} and we introduce item-specific threshold parameters τ_{jt} as follows:

$$z_{jt}^k = s, \text{ if } \tau_{s,jt} \geq z_{jt}^{*k} \geq \tau_{s+1,jt} \text{ for } s = 0, 1 \text{ and } k \in \{P, S, H, W\} \quad (8)$$

with $\tau_{0,jt} = -\infty$ and $\tau_{2,jt} = \infty$. As it is standard in the literature, we assume that the latent factors and measurement error terms are normally distributed. The expectation and variance of the measures z_{jt}^f are

$$\mu^f = \nu^f + \lambda^f \kappa^f \quad (9)$$

$$\sigma^f = (\lambda^f)^2 \psi^f + \omega^f \quad (10)$$

where κ^f and ψ^f are respectively the mean and variance of the factor $f \in \{\theta, P, S, H, W\}$ and ω^f is the variance of the residual ϵ^f for $f \in \{\theta, P, S, H, W\}$. Further sets of restrictions are required to set the location and scale of the latent factors. Specifically, we assume a so called “delta parameterisation” such that $\text{diag}(\Phi) = I, \kappa = 0, \nu = 0$ and $\text{diag}(\Sigma) = I$.⁶ This imposes restrictions on the mean and variance of the latent factors, on the intercept of z_{jt}^f and on the variance of z_{jt}^f .

5.2 The outcome equations

Our objective is twofold. First, we want to study how the treatment affects literacy skills and other latent factors which are important drivers of skill formation. Second, we want to

⁶ A popular alternative is the “theta parameterisation”, which assumes $\text{diag}(\Phi) = I, \kappa = 0, \nu = 0$ and $\text{diag}(\Omega) = I$. Whereas both methods impose restrictions on the mean and variance of the latent factors, and on the intercept of z_{jt}^f , the theta parameterisation imposes an assumption on the *residual* variance of z_{jt}^f .

Table 2: Factors and measurements related to each factor

Factor	Measure	Variable type
Literacy skills (θ_t)	Test score: correct letters per minute	Continuous
Literacy skills (θ_t)	Test score: phonetic awareness	Continuous
Literacy skills (θ_t)	Test score: familiar words fluency	Continuous
Literacy skills (θ_t)	Test score: unfamiliar words fluency	Continuous
Literacy skills (θ_t)	Test score: oral reading fluency	Continuous
Literacy skills (θ_t)	Test score: reading comprehension	Continuous
Literacy skills (θ_t)	Test score: listening comprehension	Continuous
Parental investments (P_t)	Do you have any reading books at home?	Binary
Parental investments (P_t)	Does someone read aloud to you at home?	Binary
Parental investments (P_t)	Do you practice reading aloud to someone at home?	Binary
Schooling investments (S_t)	Does your teacher practice letter sounds with you?	Binary
Schooling investments (S_t)	Do you practice reading aloud to teacher or students?	Binary
Schooling investments (S_t)	Teacher: did students read aloud in class?	Binary
Schooling investments (S_t)	Do you ever practice silent reading in school?	Binary
Schooling investments (S_t)	Does your teacher practice unfamiliar words?	Binary
Schooling investments (S_t)	Do you have a library at your school?	Binary
Schooling investments (S_t)	Do you take books from school home?	Binary
Schooling investments (S_t)	Teacher: did students have to retell a story they read?	Binary
Household background (H_t)	Does your family speak mostly English at home?	Binary
Household background (H_t)	Does your mom speak mostly English at home?	Binary
Household background (H_t)	Does your father speak mostly English at home?	Binary
Wealth (W_t)	Do you watch television at home?	Binary
Wealth (W_t)	Do you listen to radio at home?	Binary
Wealth (W_t)	Do you have electricity at home?	Binary
Wealth (W_t)	Do you have a refrigerator at home?	Binary
Wealth (W_t)	Do you have a bicycle at home?	Binary
Wealth (W_t)	Do you have a motorbike at home?	Binary
Wealth (W_t)	Do you have a car at home?	Binary
Wealth (W_t)	Does anyone in your family have a phone?	Binary

understand to what extent the treatment effects on literacy skills are mediated by improvements in the latent factors which drive skill formation. Thus, we first estimate for each child i the following set of equations:

$$\theta_i = \alpha^\theta + \gamma_1^\theta F_i + \gamma_2^\theta L_i + \gamma_3^\theta \mathbf{X}_i + v_i \quad (11a)$$

$$k_i = \alpha^k + \gamma_1^k F_i + \gamma_2^k L_i + \gamma_3^k \mathbf{X}_i + v_i, \quad k \in \{\theta, P, S, H, W\} \quad (11b)$$

where F_i is a dummy taking value one if child i belongs to the full treatment group; L_i is a dummy taking value one if child i is in the light treatment group; and $\mathbf{X}_i$ is a vector of covariates including age, grade and gender. Moreover, α^k is the intercept for factor k , which we normalize to zero ($\alpha^k = 0$); $\gamma_1^k, \gamma_2^k, \gamma_3^k$ are the coefficients capturing, respectively, the effect of full treatment, light treatment, and covariates on factor k , and v_i is the error term. We

compute clustered standard errors by bootstrap (see Section 5.4). Equation (11) can be viewed as the “factor-model” version of equation (1), in which the measures of literacy skills, parental investments and schooling investments are replaced by their corresponding latent factors.

Next, we augment equation (11) for literacy skills by allowing the latent factors for parental investment, schooling investment, household background and wealth to influence literacy skills. Thus, our structural equation for literacy skills can be written as:

$$\theta_i = \alpha^\theta + \beta^P P_i + \beta^S S_i + \beta^H H_i + \beta^W W_i + \gamma_1^\theta F_i + \gamma_2^\theta L_i + \gamma_3^\theta \mathbf{X}_i + \xi_i. \quad (12)$$

Here β^P , β^S , β^H , and β^W are respectively the effect on literacy skills of increasing parental investment, schooling investment, household background and wealth by one unit; γ_1^θ and γ_2^θ are the direct effects of respectively the full and light treatment which are not mediated through the latent factors. These can be viewed as other (unobserved) factors beyond parental investment, schooling investment, household background and wealth which are affected by the treatment and have a direct effect on literacy skills. We normalize $\alpha^\theta = 0$ in the estimation.

5.3 Mediation analysis

To estimate how much each latent factor contributed to the increase in literacy skills we use the following decomposition, which could be thought of as a three-step approach:

1. **Estimate the structural equation for literacy skills:** First, we estimate equation (12):

$$\theta_i = \alpha^\theta + \beta^P P_i + \beta^S S_i + \beta^H H_i + \beta^W W_i + \gamma_1^\theta F_i + \gamma_2^\theta L_i + \gamma_3^\theta \mathbf{X}_i + \xi_i. \quad (13)$$

where we normalize $\alpha^\theta = 0$.

2. **Estimate the base model for literacy skills:** We estimate the “base” model for literacy skills in equation (11a) that includes only the treatment and covariates:

$$\theta_i = \alpha^{\theta, \text{base}} + \gamma_1^{\theta, \text{base}} F_i + \gamma_2^{\theta, \text{base}} L_i + \gamma_3^{\theta, \text{base}} \mathbf{X}_i + v_i. \quad (14)$$

3. **Estimate auxiliary models for the mediators:** For each latent factor $k \in \{P, S, H, W\}$, we estimate equation (11b):

$$k_i = \alpha^k + \gamma_1^k F_i + \gamma_2^k L_i + \gamma_3^k \mathbf{X}_i + \chi_i. \quad (15)$$

The idea is that the difference between the treatment coefficients in the structural and base models for literacy skills captures the indirect effects of treatment mediated through the other latent factors. In particular, we can decompose these differences as:

$$\begin{aligned}\gamma_2^{\theta, base} - \gamma_2^\theta &= \sum_{k \in \{P, S, H, W\}} \gamma_2^k \beta^k \\ \gamma_1^{\theta, base} - \gamma_1^\theta &= \sum_{k \in \{P, S, H, W\}} \gamma_1^k \beta^k\end{aligned}\tag{16}$$

where for each mediator $k \in \{P, S, H, W\}$ the product $\gamma^k \times \beta^k$ represents the contribution of that latent factor to the treatment effect on literacy skills.

5.4 Estimation Algorithm

One common estimation procedure is to first estimate the factor model and then used the predicted latent factors to estimate the structural equation (12). However, measurement error from the first step could bias our estimates (See [Levi \(1973\)](#) for more details on the direction of the bias.) To overcome potential attenuation bias, we estimate the factor model and the structural equation for literacy skills jointly.

[Muthén \(1984\)](#) proposes a strategy to jointly estimate the regressions and factor model when items are categorical. The estimation protocol first estimates the parameters for the reduced form implied by the factor model and regression of interest. Those parameters are functions of the structural parameters in the factor model and regression of interest. Letting σ denote the reduced form parameters (as in [Muthén \(1984\)](#)) and ζ denote the structural parameters, we thus have that $\sigma = g(\zeta)$ for a known function $g(\cdot)$. Given the normality assumption on the error terms ϵ in equation (8), the reduced form parameters collected in σ can be estimated by maximum likelihood methods. Let $\hat{\sigma}$ denote the estimates obtained in this initial stage. Once the estimates $\hat{\sigma}$ are obtained, the procedure fits the structural parameters using a minimum distance estimator based on the following objective function:

$$F_{WLS} = [g(\zeta) - \hat{\sigma}]' \mathbf{W}^{-1} [g(\zeta) - \hat{\sigma}]\tag{17}$$

for a weight matrix $\mathbf{W}$, to be minimized with respect to ζ . [Muthén \(1984\)](#) suggests using as $\mathbf{W}$ a consistent estimator for the asymptotic covariance matrix of $\hat{\sigma}$. The corresponding estimator for ζ is referred to as the Weighted Least Squares (WLS) estimator. As the number of observed variables increases, the weight matrix rapidly grows in size, and we use an alternative matrix that is computationally more tractable and often better performing statistically, $\mathbf{W}_D =$

$diag(\tilde{\mathbf{V}})$, that only incorporates diagonal elements of the full weight matrix in the fit function. We refer to this estimation technique as Diagonally Weighted Least Squares. The estimation is carried out through the R Lavaan package (Rosseel, 2012).

Once we have estimated the structural equation in (12) and the decomposition of treatment effects in (16), we construct confidence intervals and p-values by bootstrapping 500 samples of the original data at the cluster level. There are multiple bootstrap methods to form confidence intervals. A popular and simple method is the percentile interval, which is based on the quantiles of the bootstrap distribution (Hansen, 2022). For any $0 < \alpha < 1$ we can calculate the empirical quantile q_α^* of the bootstrap estimates, which is the number such that $n\alpha$ bootstrap estimates are smaller than q_α^* . The percentile bootstrap confidence interval is:

$$C^{PC} = \left[q_{\frac{\alpha}{2}}^*, q_{\frac{1-\alpha}{2}}^* \right] \quad (18)$$

In our case, $B = 500$, and we will report the C^{PC} for $\alpha = 0.05$ in the tables. Additionally, use 90% and 99% confidence intervals to assess the significance level of our estimates.

6 Results

In this Section we present the results from the estimation of the structural model from Section 5. In the main text we present these results for the final assessment. The results of the midterm assessment are reported in Appendix C.

6.1 Measurement system

Table 3 reports the estimated factor loadings for the measurement system defined in equations (3)-(7). As we specified a dedicated measurement system, each measure loads on only one factor. Overall, the results provide strong support for the proposed latent-factor structure. With few exceptions, all indicators load positively and significantly on their intended latent construct, suggesting that the selected measures capture common underlying dimensions of children’s learning environments and reading achievement.

The literacy factor is particularly well identified. All seven reading assessment components exhibit large and precisely estimated loadings, with oral reading fluency, unfamiliar-word fluency, familiar-word fluency, and correct letters per minute displaying the strongest associations with the latent literacy construct. Reading and listening comprehension also load positively and significantly on the factor, indicating that the latent achievement measure captures both foundational decoding skills and broader comprehension abilities. The parental-investment factor is

similarly well measured, with all three indicators exhibiting large and highly significant loadings. In particular, reading aloud at home and practicing reading with family members emerge as especially informative measures of the latent home-learning environment.

The schooling-investment factor is identified through a broader set of classroom and school-level indicators. Most loadings are positive, statistically significant, and of comparable magnitude, suggesting that the factor captures a common dimension of exposure to literacy-oriented instructional practices. Indicators related to reading aloud, letter-sound instruction, unfamiliar-word practice, access to school libraries, and borrowing books all load strongly on the factor. Two teacher-reported classroom activities (reading aloud in class and story retelling) display weaker and less precisely estimated loadings, although their estimated coefficients remain positive. Finally, the household-background and wealth factors are both strongly identified. Language-use indicators exhibit large loadings on the background factor, while ownership of household assets and access to infrastructure load positively and significantly on the wealth factor. Taken together, these results suggest that the latent-factor specification provides a coherent representation of the main dimensions of children's home and school environments targeted by the intervention.

6.2 Treatment effects

Next, we present the treatment effects on the latent factors entering the literacy production function. These estimates are obtained from the reduced-form structural system, jointly estimated with the measurement model, in which each latent factor is regressed on treatment assignment and the baseline controls, without conditioning the literacy skill equation on the mediators themselves. As a result, the coefficients on the treatment indicators in the literacy skill equation should be interpreted as total reduced-form effects rather than as direct effects net of the mediator system.

The results in Table 4 reveal a clear contrast between the full and light intervention arms. Assignment to the full-treatment group led to large and statistically significant improvements in literacy skills, parental investments, and schooling investments, whereas the estimated effects on household background and wealth are small and statistically indistinguishable from zero. In contrast, the light intervention generated no detectable changes in any of the latent factors.

The estimated effect of the full intervention on literacy skills is substantial, amounting to nearly one standard deviation of the latent literacy factor. At the same time, the intervention produced sizeable increases in both parental investments and schooling investments, with estimated effects of 0.67 and 1.14 standard deviations, respectively. These findings are consistent with the program's objective of improving literacy outcomes by strengthening instructional practices in

Table 3: Estimates of factor loadings for the measurement system

Measure	Coefficient	Confidence interval
Literacy skills (θ_t)		
Correct letters per minute	17.05***	[14.96; 19.21]
Phonetic awareness	1.41***	[1.16; 1.65]
Familiar words fluency	19.24***	[16.88; 21.34]
Unfamiliar words fluency	28.99***	[25.68; 31.81]
Oral reading fluency	30.82***	[27.69; 33.20]
Reading comprehension	12.92***	[10.60; 15.34]
Listening comprehension	7.84***	[5.73; 10.31]
Parental investments (P_t)		
Do you have any reading books at home?	0.72***	[0.62; 0.81]
Does someone read aloud to you at home?	0.90***	[0.84; 0.97]
Do you practice reading aloud to someone at home?	0.93***	[0.86; 0.99]
Schooling investments (S_t)		
Does your teacher practice letter sounds with you?	0.65***	[0.39; 0.87]
Do you practice reading aloud to teacher or students?	0.64***	[0.45; 0.81]
Teacher: did students read aloud in class?	0.39*	[-0.03; 0.85]
Do you ever practice silent reading in school?	0.55***	[0.33; 0.76]
Does your teacher practice unfamiliar words?	0.60***	[0.40; 0.77]
Do you have a library at your school?	0.52***	[0.29; 0.73]
Do you take books from school home?	0.80***	[0.50; 1.21]
Teacher: did students have to retell a story they read?	0.27	[-0.15; 0.75]
Household background (H_t)		
Does your family speak mostly English at home?	0.58***	[0.45; 0.70]
Does your mom speak mostly English at home?	0.91***	[0.84; 0.98]
Does your father speak mostly English at home?	0.91***	[0.83; 0.99]
Wealth (W_t)		
Do you watch television at home?	0.99***	[0.95; 1.03]
Do you listen to radio at home?	0.52***	[0.38; 0.66]
Do you have electricity at home?	0.97***	[0.93; 1.01]
Do you have a refrigerator at home?	0.64***	[0.53; 0.73]
Do you have a bicycle at home?	0.52***	[0.44; 0.61]
Do you have a motorbike at home?	0.43***	[0.34; 0.53]
Do you have a car at home?	0.58***	[0.51; 0.65]
Does anyone in your family have a phone?	0.73***	[0.60; 0.84]

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: Entries are standardized factor loadings from the main SEM specification for the requested model. Confidence intervals (95%) are constructed from the cluster bootstrap CSV used for the main SEM table. Stars are based on bootstrap p-values.

schools while simultaneously encouraging reading-related activities in the home environment. The results indicate that the intervention successfully altered both dimensions of the learning environment targeted by the program.

By contrast, we find no evidence that the intervention affected predetermined household characteristics or material resources. Estimated effects on the household-background and wealth factors are small, with confidence intervals that comfortably include zero. This pattern provides a useful validity check for the latent-factor framework. Because these factors capture

relatively stable characteristics of children’s households that were not directly targeted by the intervention, the absence of significant treatment effects is consistent with the experimental design.

Taken together, the reduced-form results suggest that the gains in literacy achievement generated by the full intervention were accompanied by substantial changes in both parental and schooling investments. The next section investigates whether these improvements in children’s home and school learning environments can account for the observed treatment effects on literacy skills.

Table 4: Treatment effects on literacy skills and production function inputs

	Dependent Factor				
	Literacy skills	Parental investment	Schooling investment	Household background	Wealth
Full	0.97*** [0.65; 1.26]	0.67*** [0.35; 0.94]	1.14*** [0.66; 1.44]	0.15 [-0.34; 0.65]	0.05 [-0.38; 0.45]
Light	0.14 [-0.07; 0.37]	0.14 [-0.15; 0.48]	0.15 [-0.23; 0.56]	0.05 [-0.41; 0.56]	0.09 [-0.40; 0.51]
Controls	✓	✓	✓	✓	✓

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: Coefficients are from the reduced-form structural system in which RT, CB, HS, DU, and TE are each regressed on treatment and controls, without mediator terms in the RT equation. Confidence intervals (95%) are constructed from the cluster bootstrap CSV for this specification. Stars are based on bootstrap p-values. Controls include age, gender and grade.

6.3 Literacy skills formation

Next, we estimate the full structural system described by equations (11b)-(12), in which literacy skills depend on the latent inputs while each input is simultaneously modelled as a function of treatment assignment and control variables. Table 5 reports the resulting estimates. The coefficients in the literacy skill equation can be interpreted as the partial associations between each latent input and literacy skills, holding constant the remaining factors and treatment status. The coefficients in the remaining columns capture the effects of the intervention on the latent inputs.

The results indicate that both parental investments and schooling investments are important determinants of literacy achievement. A one-standard-deviation increase in parental investments is associated with a 0.19 standard deviation increase in literacy skills, while a comparable increase in schooling investments is associated with a 0.38 standard deviation increase in literacy skills. Both coefficients are statistically significant and precisely estimated. In contrast, the estimated effects of household background and wealth on literacy achievement are small and

statistically indistinguishable from zero once the remaining factors are taken into account. The estimates therefore suggest that variation in literacy outcomes is primarily associated with differences in children’s home reading environment and exposure to literacy-oriented instructional practices at school.

The treatment effects on the latent inputs are remarkably similar to those reported in Table 4. Assignment to the full-treatment arm generates large and statistically significant increases in both parental investments and schooling investments, while the effects on household background and wealth remain small and insignificant. The stability of these estimates reinforces the interpretation that the intervention operated primarily through the home and school learning environments targeted by the program. By contrast, the light intervention does not produce statistically significant changes in any of the latent factors. The absence of detectable effects on either literacy skills or the mediating inputs is consistent with the reduced-form evidence and suggests that the more limited intervention was insufficient to alter children’s learning environments in a meaningful way.

A comparison with the reduced-form estimates reported in Table 4 further suggests that these factors account for an important share of the intervention’s impact on literacy skills. In the reduced-form specification, the estimated effect of the full intervention on literacy achievement is 0.97 standard deviations. After conditioning on the latent inputs, the corresponding coefficient falls to 0.42 standard deviations. Thus, more than half of the overall treatment effect is accounted for by changes in the measured mediating factors. At the same time, the coefficient on the full-treatment indicator remains positive and statistically significant, indicating that the latent factors included in the model do not fully explain the intervention’s impact on literacy achievement. This residual effect may reflect additional mechanisms affected by the intervention but not captured by the observed measures, measurement error in the mediating factors, or departures from the assumed production function. In the next section, we formally quantify the contribution of each mediator using the decomposition discussed in Section 5.3.

6.4 Mediation analysis

Table 6 presents the mediation analysis implied by the structural model. The indirect effects are computed as the product of the treatment effect on each latent input and the corresponding coefficient in the literacy production function. The decomposition therefore partitions the total treatment effect into a direct component and a set of indirect effects operating through the latent factors included in the model.

Consistent with the preceding results, the decomposition reveals substantial mediation of the full-treatment effect through parental and schooling investments. The total effect of the full

Table 5: Estimates of the literacy skill production function and treatment effects on inputs

	Dependent Factor				
	Literacy skills	Parental investment (P_t)	Schooling investment (S_t)	Household background (H_t)	Wealth (W_t)
Full	0.42** [0.09; 0.78]	0.68*** [0.39; 0.98]	1.08*** [0.57; 1.46]	0.17 [-0.28; 0.75]	0.07 [-0.37; 0.51]
Light	0.06 [-0.23; 0.31]	0.17 [-0.15; 0.51]	0.16 [-0.30; 0.61]	0.08 [-0.39; 0.60]	0.10 [-0.37; 0.56]
P_t	0.19** [0.06; 0.32]				
S_t	0.38*** [0.18; 0.66]				
H_t	0.01 [-0.10; 0.12]				
W_t	-0.05 [-0.14; 0.04]				
Controls	✓	✓	✓	✓	✓

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: Coefficients are from the structural system only. Confidence intervals (95%) are constructed from the cluster bootstrap CSV for this specification. Stars are based on bootstrap p-values. Controls include age, gender and grade.

intervention on literacy skills is estimated at 0.97 standard deviations, of which 0.55 standard deviations are explained by changes in the measured mediating factors. The largest contribution arises from schooling investments, which account for 0.42 standard deviations of the treatment effect. Improvements in parental investments contribute an additional 0.13 standard deviations. By contrast, household background and wealth make virtually no contribution to the overall treatment effect, reflecting both the absence of treatment effects on these factors and their limited association with literacy achievement in the production function estimates.

Expressed as shares of the total treatment effect, the measured mediators account for approximately 57 percent of the impact of the full intervention on literacy skills. Schooling investments alone explain around 43 percent of the total effect, while parental investments account for a further 14 percent. These findings suggest that improvements in instructional practices and school learning environments constitute the primary mechanism through which EGRA Plus affected literacy outcomes, with increased parental engagement providing an additional but smaller contribution.

At the same time, a sizeable direct effect of the full intervention remains. Even after accounting for all measured mediators, the estimated direct effect is 0.42 standard deviations and remains statistically significant. As discussed above, this residual component may reflect mechanisms not captured by the available indicators, measurement error in the latent factors, or limitations of the assumed production function. Nevertheless, the results indicate that a substantial share of the intervention's impact can be attributed to the two channels most directly targeted by the

program: parental support for reading and literacy-oriented schooling investments.

Finally, the mediation analysis confirms the absence of meaningful effects for the light intervention. Neither the total effect nor any of the indirect effects are statistically distinguishable from zero. The lack of mediated effects is consistent with the earlier finding that the light-treatment arm did not significantly alter either literacy skills or the latent inputs entering the production function.

Table 6: Mediation analysis for literacy skills

	Full	Light
Total treatment effect	0.97*** [0.69; 1.23]	0.15 [-0.07; 0.38]
Direct effect of the treatment	0.42** [0.08; 0.76]	0.06 [-0.21; 0.32]
<i>Mediators: Indirect effect via</i> Parental investment	0.13** [0.03; 0.25]	0.03 [-0.03; 0.11]
Schooling Investment	0.42*** [0.17; 0.80]	0.06 [-0.10; 0.29]
Household background	0.00 [-0.03; 0.04]	0.00 [-0.03; 0.04]
Wealth	-0.00 [-0.04; 0.02]	-0.00 [-0.04; 0.03]

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: The direct, indirect, and total effects are taken from the SEM mediation specification for the requested model. Confidence intervals (95%) are constructed from the cluster bootstrap CSV for this specification. Controls include age, gender and grade.

6.5 Robustness checks and heterogeneity analysis

Robustness checks. We perform several robustness checks for the estimation of our structural model defined by equations (11b)-(12). The first robustness exercise augments the structural system with baseline school-level averages of the latent factors and literacy skills. These controls are intended to capture pre-existing differences across schools that may not be fully accounted for by individual-level covariates. This is a demanding robustness check because it absorbs a substantial amount of pre-program cross-school variation. Nevertheless, the results are virtually unchanged relative to the baseline specification (Appendix Table D4). The estimated effects of parental and schooling investments on literacy achievement remain stable, and the intervention continues to generate large and statistically significant improvements in both mediators. This finding is particularly important, because it suggests that the estimated mediation pattern is not simply reflecting pre-existing school differences.

Second, in Appendix Table D5 we estimate a sequence of models in which only one mediator enters the literacy equation at a time. This specification provides a transparent view of the relative importance of each mediator by including only one channel at a time in the literacy skills equation, while at the same time allowing to assess whether the relative importance of the mediating channels is sensitive to potential correlations among the latent factors. The results confirm the central role of parental and schooling investments. When either of these mediators is included individually, the estimated treatment effect on literacy skills declines substantially relative to the reduced-form effect, indicating that each channel explains a meaningful share of the intervention's impact. In contrast, including household background or wealth alone has almost no effect on the estimated treatment coefficient, which remains essentially identical to the reduced-form estimate. This pattern reinforces the conclusion that the intervention operated primarily through changes in children's home and school learning environments.

Finally, Appendix Table D6 reports estimates from a two-step procedure in which the latent factors are first recovered from the measurement model and then incorporated into the structural equations in a second stage. Although the two-step approach yields somewhat larger coefficients on parental and schooling investments and a correspondingly smaller direct treatment effect, the qualitative conclusions remain unchanged. The full intervention continues to have large positive effects on parental and schooling investments, both mediators remain positively associated with literacy achievement, and household background and wealth remain statistically insignificant. The similarity between the joint SEM estimates and the two-step results suggests that the substantive findings are not driven by the particular estimation strategy adopted in the main specification.

Overall, the findings are highly consistent with the baseline model. Across all specifications, the full intervention continues to generate substantial improvements in parental and schooling investments, while the light intervention remains statistically indistinguishable from the control group. Similarly, parental and schooling investments remain the only factors that exhibit a robust association with literacy achievement, whereas household background and wealth continue to play a limited role once the remaining inputs are taken into account.

Heterogeneity analysis Appendix D.2 examine whether the estimated literacy production function varies across student and school characteristics. Table D7 reports separate estimates for female and male students, while Table D8 estimates the model separately for schools with baseline literacy performance above and below the sample median.

The results provide little evidence of meaningful heterogeneity across either dimension. The estimated effects of parental and schooling investments on literacy achievement are remarkably similar across gender groups, and the corresponding difference estimates are uniformly small

and statistically insignificant. Likewise, the estimated coefficients obtained for schools with above- and below-median baseline literacy levels are broadly comparable, with no statistically significant differences across groups. In both cases, schooling and parental investments remain the primary factors associated with literacy achievement, while household background and wealth continue to exhibit limited explanatory power once the remaining factors are taken into account.

Taken together, these findings suggest that the mechanisms identified in the main analysis operate similarly across students and schools. The evidence therefore supports the interpretation that the estimated production function and the role of the mediating factors are broadly stable across the subpopulations considered.

6.6 Numeracy skills

Although EGRA Plus was explicitly designed to improve reading outcomes, the endline survey also collected information on a range of mathematics competencies. These data provide an opportunity to examine whether the intervention generated spillover effects on learning outcomes beyond literacy or, conversely, whether gains in reading achievement came at the expense of performance in other academic domains. Thus, in this section, we replicate the structural analysis presented for literacy skills using a latent measure of mathematics achievement constructed from the math assessment items. Appendix Table D9 reports the corresponding measurement system.

Table 7 reports reduced-form treatment effects on numeracy skills and the latent factors entering the production function. The results provide little evidence that the intervention affected mathematics achievement. The estimated effect of the full intervention on the latent mathematics factor is positive but statistically insignificant, while the light intervention has no detectable effect. At the same time, the intervention continues to generate large improvements in parental and schooling investments, mirroring the pattern observed in the literacy analysis. Thus, the intervention successfully altered the home and school learning environments targeted by the program, but these changes do not appear to translate into measurable gains in mathematics achievement.

Table 8 reports estimates of the mathematics production function. In contrast to the literacy results, the associations between the latent inputs and mathematics skills are generally weaker and less precisely estimated. Parental investments are positively associated with mathematics achievement, although the coefficient is only marginally significant, while the effects of schooling investments, household background, and wealth are not statistically distinguishable from zero. Consistent with the reduced-form results, the direct effects of both treatment arms on

Table 7: Treatment effects on math skills and production function inputs

	Dependent Factor				
	Math skills	Parental investment	Schooling investment	Household background	Wealth
Full	0.21 [-0.16; 0.56]	0.67*** [0.39; 0.99]	1.00* [-0.02; 1.44]	0.15 [-0.36; 0.64]	0.05 [-0.38; 0.49]
Light	-0.01 [-0.34; 0.32]	0.15 [-0.17; 0.49]	0.16 [-0.30; 0.63]	0.07 [-0.42; 0.60]	0.10 [-0.37; 0.56]
Controls	✓	✓	✓	✓	✓

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: Coefficients are from the reduced-form structural system in which the outcome factor and all mediator factors are each regressed on treatment and controls, without mediator terms in the outcome equation. Confidence intervals (95%) are constructed from the cluster bootstrap CSV for this specification. Stars are based on bootstrap p-values. Controls include age, gender and grade.

mathematics achievement remain insignificant once the mediating factors are included.

Table 8: Estimates of the math skill production function and treatment effects on inputs

	Dependent Factor				
	Math skills	Parental investment (P_t)	Schooling investment (S_t)	Household background (H_t)	Wealth (W_t)
Full	-0.23 [-0.67; 0.28]	0.68*** [0.39; 0.97]	0.97* [-0.12; 1.42]	0.16 [-0.32; 0.67]	0.06 [-0.35; 0.47]
Light	-0.10 [-0.45; 0.23]	0.15 [-0.21; 0.51]	0.14 [-0.28; 0.58]	0.06 [-0.37; 0.52]	0.09 [-0.33; 0.50]
P_t	0.19* [-0.04; 0.36]				
S_t	0.28 [-0.39; 0.68]				
H_t	0.06 [-0.13; 0.33]				
W_t	-0.02 [-0.23; 0.25]				
Controls	✓	✓	✓	✓	✓

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: Coefficients are from the structural system only. Confidence intervals (95%) are constructed from the cluster bootstrap CSV for this specification. Stars are based on bootstrap p-values. Controls include age, gender and grade.

Appendix Table D10 presents the corresponding mediation analysis. The estimated indirect effects through parental and schooling investments are positive, reflecting the fact that the intervention increased these inputs. However, neither the total treatment effect nor the individual mediation channels are precisely estimated at conventional significance levels. Overall, the decomposition provides little evidence that the intervention generated meaningful improvements in mathematics achievement through the mechanisms identified in the literacy analysis.

Taken together, the mathematics results suggest that the effects of EGRA Plus were largely domain specific. While the intervention substantially improved literacy skills and the learning

environments associated with reading achievement, these gains did not extend to mathematics. At the same time, the absence of negative effects on mathematics achievement indicates that improvements in literacy were not achieved through a reallocation of instructional effort away from other academic subjects. The findings therefore provide no evidence of crowding-out effects and support the interpretation that the intervention's impacts were concentrated in the domain it was explicitly designed to influence.

7 Conclusions

This paper examines the mechanisms underlying the success of the EGRA Plus Liberia literacy intervention. EGRA Plus combined teacher training, classroom coaching, instructional materials, and school-based support aimed at improving reading instruction, while also encouraging greater parental and community engagement in children's learning. Previous evaluations established that the intervention generated substantial improvements in reading achievement, particularly under the more intensive treatment arm. However, less was known about how these gains were produced. To address this question, we estimated a latent-factor model of literacy skill formation that jointly models literacy achievement and a set of underlying inputs capturing parental investments, schooling investments, household background, and household wealth.

Our findings indicate that the intervention operated primarily through changes in children's home and school learning environments. Assignment to the full-treatment arm generated large increases in both parental investments and schooling investments, while the light intervention produced no meaningful changes in either dimension. Within the literacy production function, parental and schooling investments are strongly associated with literacy achievement, whereas household background and wealth play a limited role once the remaining factors are taken into account. The mediation analysis suggests that more than half of the total treatment effect can be explained by improvements in the measured inputs, with schooling investments accounting for the largest share of the mediated effect.

These results have important implications for the design of literacy interventions. They suggest that improvements in instructional practices and learning opportunities within schools constitute a key mechanism through which literacy gains are achieved. At the same time, increased parental engagement appears to provide an additional and non-negligible contribution. The findings therefore support approaches that combine pedagogical support for teachers with efforts to strengthen children's literacy environments outside the classroom.

Finally, we find no evidence that the intervention affected mathematics achievement. Although the program substantially improved the home and school inputs associated with literacy de-

velopment, these changes did not translate into measurable gains in mathematics. Equally important, they did not generate negative spillovers on mathematics performance, suggesting that literacy gains were not achieved at the expense of other academic subjects.

Taken together, the results highlight the value of moving beyond treatment-effect estimation to study the mechanisms through which educational interventions affect learning. Understanding why successful programs work is essential for designing policies that can be adapted, scaled, and transferred across contexts.

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ONLINE APPENDIX

Supplementary material to *Unpacking succesful literacy interventions*

Pedro Carneiro, Nathalie Lahire, Emma Scandolo, and Federico Tagliati

A	Additional reduced form analysis	2
A.1	Estimated effects on reading, parental support and teaching outcomes	2
A.2	Estimated effects on math outcomes	5
A.3	Multiple hypothesis testing	7
B	Identification	8
C	The results of the midterm assessment	9
D	Additional analysis for the structural estimates	10
D.1	Robustness checks	10
D.2	Heterogeneity	13
D.3	Math skills	15

A Additional reduced form analysis

A.1 Estimated effects on reading, parental support and teaching outcomes

Table [A1](#) shows the impacts on reading, parental support and teaching outcomes from the midterm assessment estimated using equation (1) and including different sets of control variables. Table [A2](#) does the same for the final assessment.

Table A1: The impacts on reading, parental support and teaching outcomes: midterm assessment

	Control	Full – Control				Light – Control			Full – Light			Obs.
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	
<i>Panel A. Reading test scores</i>												
Correct letters per minute	-0.00	0.56*** (0.13)	0.57*** (0.06)	0.58*** (0.06)	0.45*** (0.14)	0.46*** (0.06)	0.47*** (0.06)	0.11 (0.12)	0.11** (0.05)	0.12** (0.05)	2593	
Phonemic awareness	-0.00	0.24** (0.12)	0.26*** (0.07)	0.26*** (0.07)	0.17 (0.11)	0.18*** (0.06)	0.18*** (0.06)	0.07 (0.10)	0.08 (0.06)	0.08 (0.06)	2671	
Familiar word fluency	-0.00	0.15 (0.14)	0.16** (0.07)	0.17** (0.07)	0.14 (0.14)	0.16** (0.07)	0.16** (0.07)	0.01 (0.14)	0.00 (0.07)	0.01 (0.07)	2575	
Unfamiliar word fluency	0.00	0.35** (0.13)	0.36*** (0.10)	0.37*** (0.10)	0.16 (0.15)	0.18 (0.11)	0.18 (0.11)	0.19 (0.18)	0.18* (0.10)	0.19* (0.10)	2577	
Oral reading fluency	-0.00	0.37** (0.17)	0.36*** (0.08)	0.37*** (0.08)	0.26* (0.14)	0.26*** (0.07)	0.26*** (0.07)	0.11 (0.16)	0.10 (0.08)	0.11 (0.08)	2530	
Reading comprehension	0.00	0.28 (0.18)	0.29*** (0.10)	0.29*** (0.10)	0.15 (0.15)	0.16* (0.09)	0.16* (0.09)	0.13 (0.16)	0.13 (0.09)	0.13 (0.09)	2530	
Listening comprehension	-0.00	0.31 (0.19)	0.32*** (0.10)	0.32*** (0.10)	0.28 (0.18)	0.29*** (0.09)	0.29*** (0.09)	0.03 (0.13)	0.02 (0.07)	0.02 (0.07)	2592	
<i>Panel B. Parental support and schooling</i>												
Has reading books at home	0.50	0.28*** (0.05)	0.29*** (0.04)	0.29*** (0.04)	0.00 (0.06)	0.00 (0.04)	0.00 (0.04)	0.28*** (0.05)	0.28*** (0.04)	0.28*** (0.04)	2511	
Being read aloud at home	0.53	0.11** (0.05)	0.13*** (0.04)	0.12*** (0.04)	0.09 (0.06)	0.10*** (0.03)	0.10*** (0.03)	0.02 (0.05)	0.03 (0.03)	0.03 (0.02)	2504	
Reads aloud to someone at home	0.59	0.11* (0.06)	0.13*** (0.04)	0.13*** (0.04)	0.02 (0.06)	0.03 (0.04)	0.03 (0.04)	0.09* (0.05)	0.10*** (0.03)	0.10*** (0.03)	2504	
Has library at school	0.14	0.22*** (0.07)	0.18*** (0.05)	0.18*** (0.05)	0.12 (0.08)	0.10** (0.05)	0.10** (0.05)	0.09 (0.08)	0.08 (0.05)	0.08 (0.05)	2451	
Has books at school to take home	0.36	0.47*** (0.07)	0.47*** (0.07)	0.47*** (0.07)	0.06 (0.08)	0.07 (0.06)	0.07 (0.06)	0.41*** (0.07)	0.41*** (0.06)	0.40*** (0.06)	2489	
Teacher practices letter sounds	0.54	0.40*** (0.08)	0.37*** (0.06)	0.37*** (0.06)	0.15 (0.10)	0.13** (0.05)	0.13** (0.05)	0.26*** (0.06)	0.24*** (0.04)	0.24*** (0.04)	2463	
Reads aloud to teacher/classmates	0.86	0.03 (0.03)	0.03 (0.02)	0.03 (0.02)	0.01 (0.03)	0.01 (0.02)	0.01 (0.02)	0.02 (0.03)	0.02 (0.02)	0.02 (0.02)	2486	
Teacher reads aloud	0.97	-0.02 (0.03)	-0.02 (0.01)	-0.02 (0.01)	-0.01 (0.03)	-0.01 (0.01)	-0.01 (0.01)	-0.01 (0.04)	-0.01 (0.01)	-0.01 (0.01)	2502	
Teacher sounds unfamiliar words	0.45	0.30*** (0.08)	0.27*** (0.05)	0.27*** (0.05)	0.05 (0.09)	0.01 (0.04)	0.01 (0.05)	0.25*** (0.07)	0.26*** (0.04)	0.26*** (0.04)	2454	
District fixed effects			✓	✓		✓	✓		✓	✓		
Individual controls												

Notes. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Column (1) shows the average of the dependent variable in the control group in the midterm assessment. Columns (2)–(4) and columns (5)–(7) present, respectively, estimates of β^F and β^L from equation (1) using different sets of control variables. Columns (8)–(10) present estimates of $\beta^F - \beta^L$. Individual controls include a continuous measure of age and gender and grade dummies. Standard errors are clustered at the schooling cluster level and are presented in parentheses.

Table A2: The impacts on reading, parental support and teaching outcomes: final assessment

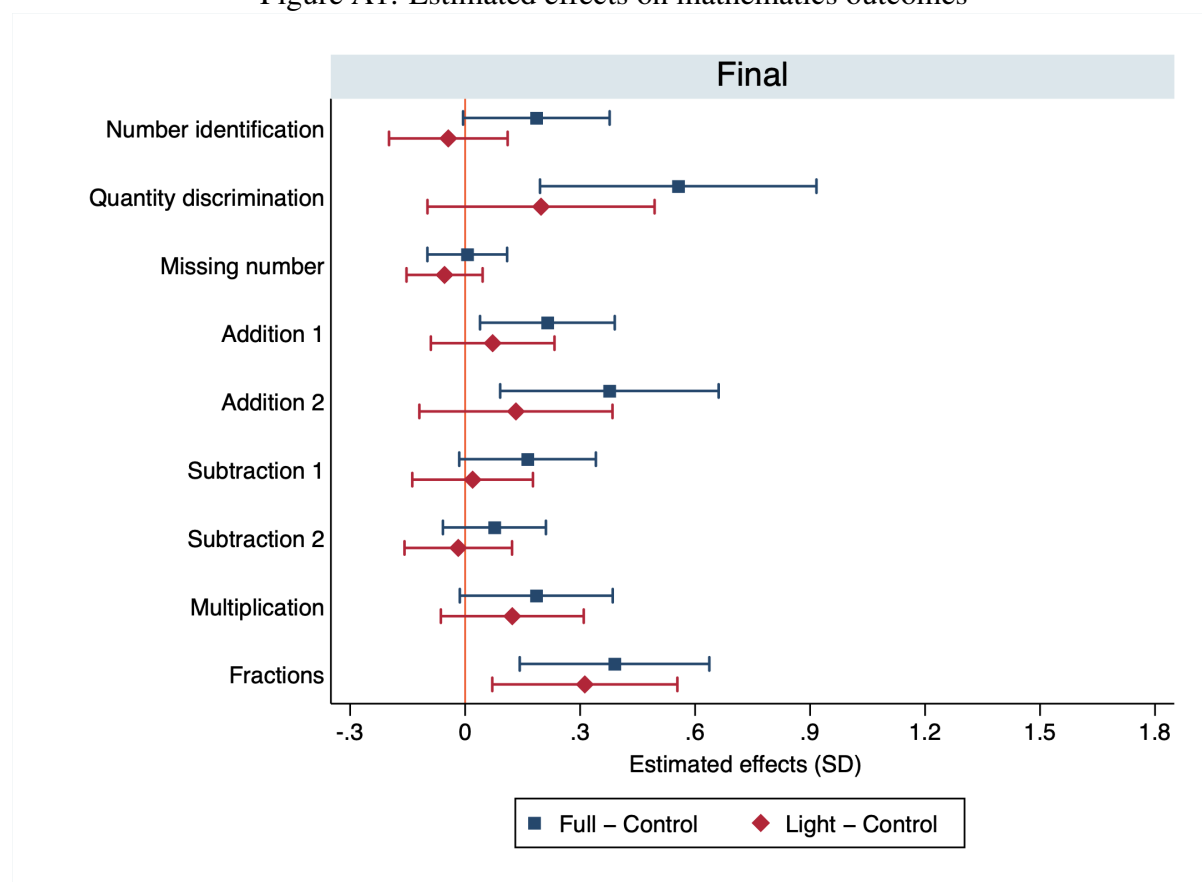
	Control mean	Full – Control		Light – Control		Full – Light		Obs.			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
<i>Panel A. Reading test scores</i>											
Correct letters per minute	-0.00	0.70*** (0.14)	0.72*** (0.14)	0.72*** (0.14)	0.25** (0.12)	0.28** (0.12)	0.27** (0.12)	0.45*** (0.13)	0.44*** (0.10)	0.45*** (0.10)	2343
Phonemic awareness	-0.00	0.57*** (0.16)	0.56*** (0.11)	0.55*** (0.11)	0.20 (0.12)	0.21** (0.09)	0.19** (0.09)	0.37** (0.16)	0.35*** (0.09)	0.35*** (0.09)	2508
Familiar word fluency	0.00	0.93*** (0.20)	0.95*** (0.19)	0.94*** (0.19)	0.07 (0.13)	0.12 (0.14)	0.11 (0.14)	0.86*** (0.18)	0.83*** (0.16)	0.83*** (0.16)	2307
Unfamiliar word fluency	0.00	1.31*** (0.37)	1.31*** (0.30)	1.30*** (0.30)	0.07 (0.15)	0.04 (0.20)	0.03 (0.20)	1.24*** (0.37)	1.27*** (0.26)	1.27*** (0.26)	2335
Oral reading fluency	0.00	0.97*** (0.22)	1.04*** (0.21)	1.02*** (0.21)	0.13 (0.13)	0.22 (0.15)	0.21 (0.15)	0.84*** (0.23)	0.82*** (0.19)	0.82*** (0.18)	2200
Reading comprehension	-0.00	0.86*** (0.17)	0.90*** (0.17)	0.89*** (0.17)	0.11 (0.13)	0.18 (0.13)	0.17 (0.13)	0.75*** (0.18)	0.72*** (0.15)	0.72*** (0.14)	2200
Listening comprehension	-0.00	0.45*** (0.10)	0.46*** (0.10)	0.46*** (0.10)	0.10 (0.11)	0.12 (0.08)	0.12 (0.08)	0.35*** (0.09)	0.34*** (0.08)	0.34*** (0.08)	2448
<i>Panel B. Parental support and schooling</i>											
Has reading books at home	0.51	0.32*** (0.05)	0.32*** (0.05)	0.32*** (0.05)	0.05 (0.07)	0.04 (0.05)	0.04 (0.05)	0.27*** (0.05)	0.28*** (0.04)	0.28*** (0.04)	2423
Being read aloud at home	0.57	0.15** (0.06)	0.15*** (0.05)	0.15*** (0.05)	0.04 (0.06)	0.05 (0.05)	0.04 (0.05)	0.10** (0.05)	0.11*** (0.04)	0.11*** (0.04)	2488
Reads aloud to someone at home	0.55	0.20*** (0.07)	0.21*** (0.05)	0.21*** (0.06)	0.07 (0.06)	0.07 (0.05)	0.07 (0.05)	0.13*** (0.05)	0.14*** (0.04)	0.14*** (0.04)	2416
Has library at school	0.20	0.21** (0.08)	0.21*** (0.07)	0.21*** (0.07)	0.07 (0.08)	0.09 (0.07)	0.09 (0.07)	0.14 (0.09)	0.12* (0.06)	0.12* (0.06)	2397
Has books at school to take home	0.39	0.45*** (0.07)	0.46*** (0.05)	0.46*** (0.05)	0.11 (0.08)	0.11** (0.05)	0.11** (0.05)	0.34*** (0.06)	0.35*** (0.04)	0.34*** (0.04)	2404
Teacher practices letter sounds	0.56	0.41*** (0.09)	0.39*** (0.06)	0.39*** (0.06)	0.06 (0.12)	0.07 (0.07)	0.07 (0.07)	0.35*** (0.09)	0.32*** (0.06)	0.32*** (0.06)	2348
Reads aloud to teacher/classmates	0.89	0.03 (0.03)	0.02 (0.02)	0.02 (0.02)	-0.00 (0.03)	-0.00 (0.02)	-0.00 (0.02)	0.03 (0.03)	0.03 (0.02)	0.02 (0.02)	2397
Teacher reads aloud	0.98	0.01 (0.01)	0.01 (0.01)	0.01 (0.01)	0.00 (0.01)	0.00 (0.01)	0.00 (0.01)	0.01 (0.01)	0.01 (0.01)	0.01 (0.01)	2413
Teacher sounds unfamiliar words	0.41	0.10 (0.10)	0.09 (0.10)	0.09 (0.10)	-0.05 (0.11)	-0.05 (0.10)	-0.05 (0.10)	0.15* (0.08)	0.14** (0.06)	0.14** (0.06)	1470
District fixed effects			✓	✓		✓	✓		✓	✓	
Individual controls											

Notes. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Column (1) shows the average of the dependent variable in the control group in the final assessment. Columns (2)–(4) and columns (5)–(7) present, respectively, estimates of β^F and β^L from equation (1) using different sets of control variables. Columns (8)–(10) present estimates of $\beta^F - \beta^L$. Individual controls include a continuous measure of age and gender and grade dummies. Standard errors are clustered at the schooling cluster level and are presented in parentheses.

A.2 Estimated effects on math outcomes

Figure A1 presents reduced-form estimates of the effects of the intervention on each mathematics competency measured in the final assessment. The figure reports point estimates and 90 percent confidence intervals from equation (1), controlling for age, gender, and grade and including district fixed effects. The underlying estimates correspond to columns (4) and (7) of Table A3, which also reports results from alternative specifications without control variables and with district fixed effects only.

Figure A1: Estimated effects on mathematics outcomes



Notes. The figures show the point estimates and 90% confidence intervals of the effect of the full and light treatment on the nine mathematics domains examined in the EGRA assessment. Mathematics outcomes are standardized to have mean zero and unit standard deviation in the control group within each survey round. Estimates are based on equation (1) and include a continuous measure of age, gender and grade dummies and district fixed effects as controls. Confidence intervals are calculated based on standard errors clustered at the schooling cluster level.

Table A3: The impacts on mathematics outcomes: final assessment

	Control mean	Full – Control				Light – Control			Full – Light			Obs.
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	
Number identification	-0.00	0.14 (0.13)	0.18 (0.11)	0.19 (0.11)	-0.09 (0.13)	-0.05 (0.09)	-0.04 (0.09)	0.24* (0.13)	0.23** (0.12)	0.23* (0.11)	2416	
Quantity discrimination	-0.00	0.52* (0.26)	0.56** (0.21)	0.56** (0.21)	0.20 (0.17)	0.20 (0.18)	0.20 (0.18)	0.32 (0.28)	0.35* (0.21)	0.36* (0.21)	1971	
Missing number	-0.00	0.03 (0.10)	0.00 (0.06)	0.01 (0.06)	-0.01 (0.10)	-0.06 (0.06)	-0.05 (0.06)	0.04 (0.09)	0.06 (0.07)	0.06 (0.07)	2421	
Addition 1	0.00	0.21 (0.12)	0.21** (0.10)	0.21** (0.10)	0.06 (0.11)	0.07 (0.10)	0.07 (0.10)	0.15 (0.13)	0.14 (0.09)	0.14 (0.09)	2417	
Addition 2	-0.00	0.35 (0.24)	0.38** (0.17)	0.38** (0.17)	0.08 (0.15)	0.13 (0.15)	0.13 (0.15)	0.27 (0.25)	0.25* (0.13)	0.24* (0.13)	2415	
Subtraction 1	-0.00	0.15 (0.16)	0.16 (0.11)	0.16 (0.11)	-0.00 (0.14)	0.01 (0.09)	0.02 (0.09)	0.16 (0.15)	0.14 (0.10)	0.14 (0.10)	2415	
Subtraction 2	-0.00	0.08 (0.09)	0.08 (0.08)	0.08 (0.08)	-0.04 (0.10)	-0.01 (0.08)	-0.02 (0.08)	0.11 (0.09)	0.09 (0.07)	0.09 (0.07)	2409	
Multiplication	-0.00	0.18 (0.14)	0.18 (0.12)	0.19 (0.12)	0.09 (0.11)	0.12 (0.11)	0.12 (0.11)	0.10 (0.16)	0.06 (0.11)	0.06 (0.11)	2410	
Fractions	0.00	0.41** (0.19)	0.39** (0.15)	0.39** (0.15)	0.33** (0.16)	0.31** (0.14)	0.31** (0.14)	0.08 (0.22)	0.08 (0.16)	0.08 (0.16)	2415	
District fixed effects			✓	✓		✓	✓		✓	✓		
Individual controls				✓								

Notes. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Column (1) shows the average of the dependent variable in the control group in the final assessment. Columns (2)–(4) and columns (5)–(7) present, respectively, estimates of β^F and β^L from equation (1) using different sets of control variables. Columns (8)–(10) present estimates of $\beta^F - \beta^L$. Individual controls include a continuous measure of age and gender and grade dummies. Standard errors are clustered at the schooling cluster level and are presented in parentheses.

A.3 Multiple hypothesis testing

TO BE ADDED

B Identification

TO BE ADDED

C The results of the midterm assessment

TO BE ADDED

D Additional analysis for the structural estimates

D.1 Robustness checks

This appendix presents a series of robustness checks designed to assess the sensitivity of the main results to alternative specifications and estimation procedures. Appendix Table [D4](#) reports estimates from a specification that augments the baseline model with school-level baseline averages of literacy skills and the latent inputs. These additional controls are included to account for potential pre-existing differences across schools that may not be fully captured by individual-level covariates.

Appendix Table [D5](#) examines the role of each mediating channel separately by estimating a sequence of specifications in which only one mediator enters the literacy production function at a time. The measurement system remains unchanged across specifications, allowing the contribution of each latent factor to be evaluated in isolation.

Finally, Appendix Table [D6](#) reports estimates obtained from a two-step procedure. In the first step, the latent factors are recovered from the measurement model. In the second step, the predicted factor scores are used to estimate the structural system relating literacy skills, treatment assignment, and the mediating factors. Comparing these estimates with the baseline joint SEM results provides evidence on the sensitivity of the findings to the estimation strategy.

Table D4: Estimates of the literacy skill production function and treatment effects on inputs with baseline mediator means

	Dependent Factor				
	Literacy skills (θ_t)	Parental investment (P_t)	Schooling investment (S_t)	Household background (H_t)	Wealth (W_t)
Full	0.41** [0.08; 0.76]	0.67*** [0.35; 1.00]	1.01*** [0.58; 1.36]	0.30* [-0.07; 0.64]	0.08 [-0.20; 0.37]
Light	0.03 [-0.22; 0.27]	0.16 [-0.19; 0.50]	0.14 [-0.24; 0.57]	0.07 [-0.24; 0.40]	0.04 [-0.22; 0.31]
P_t	0.19** [0.05; 0.32]				
S_t	0.40*** [0.20; 0.69]				
H_t	0.00 [-0.10; 0.11]				
W_t	-0.06 [-0.15; 0.03]				
$\bar{\theta}_0$	0.24* [-0.01; 0.52]				
$\bar{P}_0$	0.13 [-0.54; 0.74]	-0.17 [-1.02; 0.64]			
$\bar{S}_0$	-0.02 [-0.57; 0.57]		-0.51 [-1.22; 0.53]		
$\bar{H}_0$	0.04 [-0.56; 0.68]			0.43 [-0.13; 0.93]	
$\bar{W}_0$	-0.04 [-0.59; 0.49]				0.94*** [0.55; 1.25]
Controls	✓	✓	✓	✓	✓

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: This specification adds baseline school-level mean controls for the mediators and RT. Confidence intervals (95%) are constructed from the cluster bootstrap CSV for this specification. Controls include age, gender and grade.

Table D5: Estimates of the literacy skill production function: robustness check controlling for one mediator at a time

	Literacy skills			
	(1)	(2)	(3)	(4)
Full	0.76*** [0.45; 1.07]	0.48** [0.10; 0.85]	0.97*** [0.67; 1.26]	0.97*** [0.66; 1.28]
Light	0.10 [-0.10; 0.30]	0.07 [-0.21; 0.34]	0.15 [-0.05; 0.36]	0.14 [-0.05; 0.36]
P_t	0.32*** [0.24; 0.41]			
S_t		0.45*** [0.29; 0.70]		
H_t			0.07* [-0.01; 0.17]	
W_t				0.07* [-0.01; 0.15]
Controls	✓	✓	✓	✓

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: Each column keeps the full measurement system for the requested model, but only one mediator enters the RT equation at a time. Confidence intervals (95%) are constructed from the cluster bootstrap CSV for this specification. Controls include age, gender and grade.

Table D6: Two-step estimates of the literacy skill production function and treatment effects on inputs

	Dependent Factor				
	Literacy skills	Parental investment (P_t)	Schooling investment (S_t)	Household background (H_t)	Wealth (W_t)
Full	0.23* [-0.01; 0.51]	0.54*** [0.33; 0.78]	0.86*** [0.60; 1.11]	0.12 [-0.21; 0.43]	0.09 [-0.21; 0.41]
Light	0.01 [-0.18; 0.18]	0.11 [-0.11; 0.32]	0.10 [-0.13; 0.37]	0.06 [-0.23; 0.36]	0.10 [-0.23; 0.44]
P_t	0.25* [-0.02; 0.54]				
S_t	0.62*** [0.29; 0.93]				
H_t	0.05 [-0.18; 0.25]				
W_t	-0.11 [-0.27; 0.07]				
Controls	✓	✓	✓	✓	✓

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: Columns report the second-stage coefficients from the two-step SEM, with latent factors predicted from the measurement model in the first step. Confidence intervals (95%) are constructed from the cluster bootstrap CSV for this specification. Controls include age, gender and grade.

D.2 Heterogeneity

This appendix examines whether the estimated literacy production function differs across student and school characteristics. To this end, the model is estimated separately for a number of subgroups, allowing both the production-function coefficients and the treatment effects on the latent factors to vary across samples.

Appendix Table [D7](#) reports estimates separately for female and male students. This specification is used to assess whether the relationship between literacy skills and the latent inputs, as well as the effects of the intervention on these inputs, differs by gender.

Appendix Table [D8](#) reports estimates separately for schools with baseline literacy performance above and below the sample median. This specification is intended to examine whether the literacy production function and the intervention effects vary according to the initial quality of the school environment.

Table D7: Heterogeneity by gender in the estimates of the literacy skill production function

	Female (1)	Male (2)	Difference (2)-(1)
Full	0.42** [0.00; 0.83]	0.41** [0.00; 0.79]	-0.01 [-0.38; 0.33]
Light	0.05 [-0.24; 0.32]	0.04 [-0.22; 0.31]	-0.01 [-0.22; 0.18]
P_t	0.18** [0.00; 0.34]	0.18** [0.00; 0.33]	0.00 [-0.16; 0.15]
S_t	0.36*** [0.00; 0.66]	0.36*** [0.00; 0.72]	0.00 [-0.30; 0.37]
H_t	0.00 [-0.14; 0.14]	-0.00 [-0.13; 0.14]	-0.00 [-0.15; 0.16]
W_t	-0.04 [-0.16; 0.09]	-0.04 [-0.16; 0.08]	0.00 [-0.16; 0.17]
Controls	✓	✓	✓

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: The gender heterogeneity specification reports the raw gender-group coding used in the analysis data. Confidence intervals (95%) are constructed from the cluster bootstrap CSV for this specification. Controls include age and grade.

Table D8: Heterogeneity by baseline school quality in the estimates of the literacy skill production function

	Below median (1)	Above median (2)	Difference (2)-(1)
Full	0.16 [-0.38; 0.83]	0.13 [-0.50; 0.79]	-0.03 [-1.14; 1.00]
Light	-0.04 [-0.48; 0.27]	-0.05 [-0.50; 0.25]	-0.01 [-0.63; 0.61]
P_t	0.10* [-0.03; 0.37]	0.09 [-0.09; 0.37]	-0.01 [-0.39; 0.36]
S_t	0.23** [0.00; 1.02]	0.26** [0.00; 1.08]	0.03 [-0.92; 0.98]
H_t	0.02 [-0.09; 0.19]	0.03 [-0.11; 0.21]	0.01 [-0.22; 0.24]
W_t	-0.04 [-0.19; 0.02]	-0.04 [-0.19; 0.03]	-0.00 [-0.15; 0.14]
Controls	✓	✓	✓

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: The school-quality heterogeneity specification groups schools by the baseline above-median indicator. Confidence intervals (95%) are constructed from the cluster bootstrap CSV for this specification. Controls include age, gender and grade.

D.3 Math skills

Appendix Table D9 presents the measurement system used to construct the latent mathematics factor from the endline assessment items. Appendix Table D10 reports the corresponding mediation decomposition of the treatment effects into direct and indirect components operating through parental investments, schooling investments, household background, and wealth.

Table D9: Estimates of factor loadings for the measurement system for math skills

Measure	Coefficient	Confidence interval
Math skills (θ_t^M)		
Number identification	2.55***	[1.90; 3.14]
Quantity discrimination	-0.17	[-0.60; 0.11]
Missing number	-0.12**	[-0.24; -0.02]
Addition 1	3.08***	[2.52; 3.75]
Subtraction 1	2.78***	[2.37; 3.15]
Subtraction 2	1.66***	[1.25; 2.20]
Multiplication	0.58***	[0.33; 0.82]
Fractions	2.94**	[0.88; 4.78]
Parental investments (P_t)		
Do you have any reading books at home?	0.73***	[0.61; 0.85]
Does someone read aloud to you at home?	0.91***	[0.85; 0.97]
Do you practice reading aloud to someone at home?	0.91***	[0.85; 0.98]
Schooling investments (S_t)		
Does your teacher practice letter sounds with you?	0.55*	[-0.14; 0.85]
Do you practice reading aloud to teacher or students?	0.49	[-0.42; 0.69]
Teacher: did students read aloud in class?	0.30	[-0.48; 0.95]
Do you ever practice silent reading in school?	0.43	[-0.39; 0.70]
Does your teacher practice unfamiliar words?	0.54	[-0.32; 0.78]
Do you have a library at your school?	0.59	[-0.43; 0.86]
Do you take books from school home?	0.73	[-0.50; 1.09]
Teacher: did students have to retell a story they read?	0.23	[-0.48; 0.82]
Household background (H_t)		
Does your family speak mostly English at home?	0.63***	[0.47; 0.85]
Does your mom speak mostly English at home?	0.87***	[0.73; 0.96]
Does your father speak mostly English at home?	0.92***	[0.83; 1.01]
Wealth (W_t)		
Do you watch television at home?	1.01***	[0.94; 1.15]
Do you listen to radio at home?	0.52***	[0.36; 0.66]
Do you have electricity at home?	0.95***	[0.85; 1.00]
Do you have a refrigerator at home?	0.63***	[0.51; 0.72]
Do you have a bicycle at home?	0.51***	[0.39; 0.61]
Do you have a motorbike at home?	0.43***	[0.30; 0.56]
Do you have a car at home?	0.58***	[0.49; 0.67]
Does anyone in your family have a phone?	0.71***	[0.57; 0.84]

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: Entries are standardized factor loadings from the main SEM specification for the requested model. Confidence intervals (95%) are constructed from the cluster bootstrap CSV used for the main SEM table. Stars are based on bootstrap p-values.

Table D10: Mediation analysis for math skills

	Full	Light
Total	0.21 [-0.14; 0.56]	0.00 [-0.30; 0.32]
Direct	-0.24 [-0.66; 0.28]	-0.09 [-0.42; 0.20]
<i>Mediators: Indirect effect via</i>		
Indirect via CB	0.01 [-0.07; 0.10]	0.01 [-0.05; 0.10]
Indirect via HS	0.13* [-0.02; 0.29]	0.03 [-0.03; 0.13]
Indirect via DU	0.00 [-0.05; 0.06]	-0.00 [-0.07; 0.06]
Indirect via TE	0.31* [-0.05; 0.65]	0.06 [-0.11; 0.28]

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notes: The direct, indirect, and total effects are taken from the SEM mediation specification for the requested model. Confidence intervals (95%) are constructed from the cluster bootstrap CSV for this specification. Controls include age, gender and grade.